

Adaptive Sampling for Monte-Carlo Event Imagery Rendering

Yuichiro Manabe
24wm3206@student.chiba-u.jp
Chiba University
Chiba, Japan

Shigeo Morishima
shigeo@waseda.jp
Waseda University
Tokyo, Japan

Tatsuya Yatagawa
tatsuya.yatagawa@r.hit-u.ac.jp
Hitotsubashi University
Tokyo, Japan

Hiroyuki Kubo
hkubo@chiba-u.jp
Chiba University
Chiba, Japan

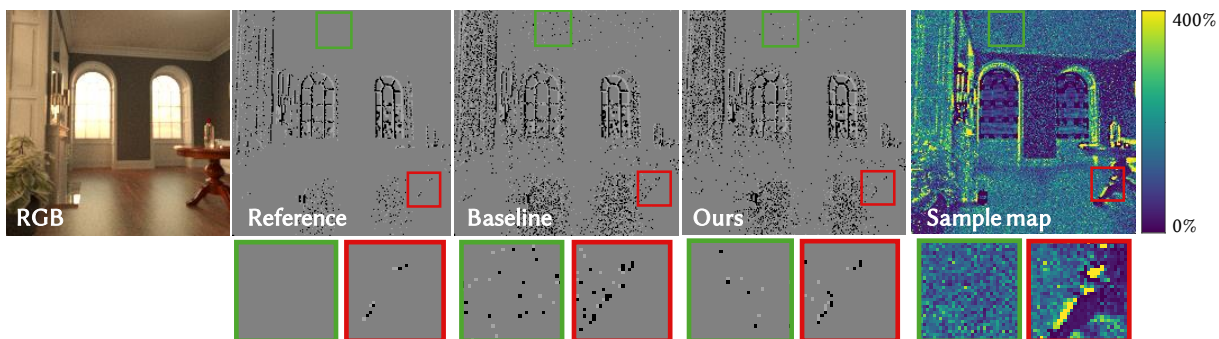


Figure 1: Our event camera simulator assigns path samples adaptively to increase the quality of resulting event imagery. The sampling map used in our method exhibits high sampling ratios at the pixels with high variance (e.g., glossy floors), and where events may more likely occur (e.g., the boundary of a table leg).

ABSTRACT

This paper presents a novel event-based camera simulation system based on physically accurate Monte Carlo path tracing with adaptive path sampling. The adaptive sampling performed in the proposed method is based on the probability of event occurrence. First, our rendering system collects logarithmic luminances rather than raw luminance based on the circuit characteristics of event-based cameras. We calculate the probability of how much different the logarithmic luminance gap is from the preset event threshold. This means how likely an event will occur at the pixel. Then, we sample paths adaptively based on the sample rate combined with a previous adaptive sampling method. We demonstrate that our method achieves higher rendering quality than the baseline approach which allocates path samples uniformly for each pixel.

CCS CONCEPTS

• **Computing methodologies** → **Ray tracing; Computational photography.**

KEYWORDS

Event-based camera, Monte-Carlo path tracing, adaptive sampling

ACM Reference Format:

Yuichiro Manabe, Tatsuya Yatagawa, Shigeo Morishima, and Hiroyuki Kubo. 2024. Adaptive Sampling for Monte-Carlo Event Imagery Rendering. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641234.3671028>

1 INTRODUCTION

The event-based camera is a device that detects temporal brightness changes above a predefined threshold as events [Lichtsteiner et al. 2008]. Owing to its asynchronous and individual event detection, event cameras have been applied to many computer vision applications [Gallego et al. 2022]. However, simulating event cameras is still challenging as previous event-based camera simulators [Hu et al. 2021; Rebecq et al. 2018] mainly focus on converting RGB video frames into events, which is problematic with noisy frames obtained by Monte Carlo path tracing. These simulators are highly sensitive to temporal noise, resulting in significant false events.

Monte Carlo path tracing is one of the most important research topics in computer graphics and has been studied for several decades. The state-of-the-art path tracing method can produce nearly noise-free graphics in a short computation time. However, events in event video frames occur in only a few pixels, making uniform computational resource allocation inefficient, as shown in Fig. 1.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Posters '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0516-8/24/07

<https://doi.org/10.1145/3641234.3671028>

2 MONTE-CARLO-BASED EVENT RENDERER

The event-based camera detects changes in logarithmic luminances as events when the change surpasses a predefined threshold. Therefore, to make the rendering process more efficient, we aim to trace adaptively more samples at pixels where false event detection is more likely to occur. In this study, we define such pixels with the following conditions.

- (i) Pixels where significant variance may still remain;
- (ii) Pixels with a higher likelihood of event occurrence.

The first condition is to reduce the variance of the Monte Carlo integration, which is needed to reduce the false event detection due to the temporal incoherence of noise patterns. The second condition is to avoid sampling paths for regions where events do not occur.

Adaptive sampling for robust event detection. To reduce the variance in Monte Carlo integration, we employ an adaptive sampling approach [Rousselle et al. 2012]. For each pixel, two sets of path samples are traced and compared using an NL-means filter. In subsequent cycles, more path samples are traced at pixels with significant discrepancies in the previous values. Our method samples the logarithmic luminance for each path, though it can integrate any adaptive sampling method that provides a sampling probability ratio map. We need to sample more paths to accurately calculate the brightness difference at a pixel and predict the occurrence of events. We model the probability of an event based on the difference in logarithmic luminance. At each pixel, the sample means and variance of brightness is computed. Assuming a normal distribution due to the central limit theorem, we define a stochastic variable X to represent the brightness difference between two time periods, a and b .

$$X \sim \mathcal{N}(|\mu_b - \mu_a|, \sigma_a^2 + \sigma_b^2), \quad (1)$$

where μ and σ are the sample means and variances at times a and b . The sampling probability for event detection is defined by the upper probability $P(X > \tau)$ with a threshold τ .

Integrating sampling maps. Two sampling maps obtained by the above processes are integrated by pixel-wise multiplication. The resulting sample map is shown on the right of Fig. 1. In this map, a pixel is more yellowish when more paths are sampled for it, while it is more purplish if fewer paths are sampled. As shown by this sample map, our method can adaptively allocate more paths to pixels with high variance and a higher likelihood of event occurrence.

3 EXPERIMENTS AND RESULTS

The results of visual and quantitative evaluations are shown in Fig. 1 and Table 1, respectively. Although we have tested our method with three test scenes, including “Two boxes,” “Living room,” and “San Miguel,” we show the rendering result for only “Living room” in Fig. 1 due to limitations of space. For the results of the other scenes and animated results, refer to the supplementary materials. In the sample map shown on the right of Fig. 1, the sample rate is clamped to fall on the range between 0 % and 400 % percent for ease of recognition of the difference in sample budgets.

Experiment setup. The quantitative comparison employs four evaluation metrics, i.e., PSNR, RMSE, F1 score, and chamfer distance (CD). PSNR and RMSE are calculated for video frames where

Table 1: Comparison in evaluation scores between the baseline and our methods.

		Living Room	San Miguel	Two Boxes
Baseline	RMSE ↓	49.0	34.7	40.9
	PSNR ↑	13.93	17.34	15.45
	F1 Score ↑	0.97	0.99	0.82
	CD ↓	37.5×10^{-5}	6.3×10^{-5}	98.4×10^{-5}
	Time(s)	42.15	46.12	15.21
Ours	RMSE ↓	44.7	39.1	39.6
	PSNR ↑	14.6	16.2	15.7
	F1 Score ↑	0.98	0.99	0.83
	CD ↓	14.5×10^{-5}	8.6×10^{-5}	92.5×10^{-5}
	Time(s)	42.87	55.16	14.87

pixel values are set as 0.0, 0.5, and 1.0 for negative, null, and positive events, respectively. On the other hand, F1 score and chamfer distance are calculated by regarding a set of events as a “point set with signs,” which are used in a previous study [Tsuiji et al. 2023] for evaluating event video qualities. In this experiment, we tested the proposed method on a computer equipped with Intel Core i9-13900K CPU (5.80 GHz, 24 cores), NVIDIA RTX A6000 GPU, and 128 GB of RAM.

Summary of results. As shown in the enlarged image parts of Fig. 1, our method succeeded in reducing noisy false event detection and in getting the results more similar to the reference, even though the visual comparison of their frames is not straightforward due to the granular nature of event videos. In particular, false event detection on the ceiling and around the table leg is significantly reduced. Furthermore, better evaluation scores in Table 1 demonstrate the effectiveness of our method. Specifically, with the same computation time, our method marked better evaluation scores than the baseline approach of uniformly sampling the same number of paths for every pixel. Thus, the proposed method is well suited to the purpose of adaptive sampling for path-tracing-based event camera simulation.

ACKNOWLEDGMENTS

This work was sponsored by the JSPS KAKENHI with grant numbers JP24K02953, JP24K00783, and JST FOREST JPMJFR206I.

REFERENCES

- Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J Davison, Jörg Conrath, Kostas Daniilidis, and Davide Scaramuzza. 2022. Event-Based Vision: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 1 (2022), 154–180. <https://doi.org/10.1109/tpami.2020.3008413>
- Yuhuang Hu, Shih-Chii Liu, and Tobi Delbruck. 2021. v2e: From Video Frames to Realistic DVS Events. 1312–1321. <https://doi.org/10.1109/cvprw53098.2021.00144>
- Patrick Lichtsteiner, Christoph Posch, and Tobi Delbruck. 2008. A 128×128 120 dB 15 μ s Latency Asynchronous Temporal Contrast Vision Sensor. *IEEE Journal of Solid-State Circuits* 43, 2 (2008), 566–576. <https://doi.org/10.1109/jssc.2007.914337>
- Henri Rebecq, Daniel Gehrig, and Davide Scaramuzza. 2018. ESIM: an open event camera simulator. In *Conference on Robot Learning*. 969–982.
- Fabrice Rousselle, Claude Knaus, and Matthias Zwicker. 2012. Adaptive Rendering with Non-Local Means Filtering. *ACM Transactions on Graphics* 31, 6 (2012), 1–11. <https://doi.org/10.1145/2366145.2366214>
- Yuta Tsuiji, Tatsuya Yatagawa, Hiroyuki Kubo, and Shigeo Morishima. 2023. Event-Based Camera Simulation Using Monte Carlo Path Tracing with Adaptive Denoising. In *Proceedings of the IEEE International Conference on Image Processing (ICIP)*. IEEE. <https://doi.org/10.1109/icip49359.2023.10222771>